EMRECAN GULAY

UX research for complex AI and agentic systems

I study what people and AI systems do together across an interaction, then turn those findings into inspectable prototypes and research directions.

PARTICIPANT STUDY

3 workshops · 22 qualitative researchers

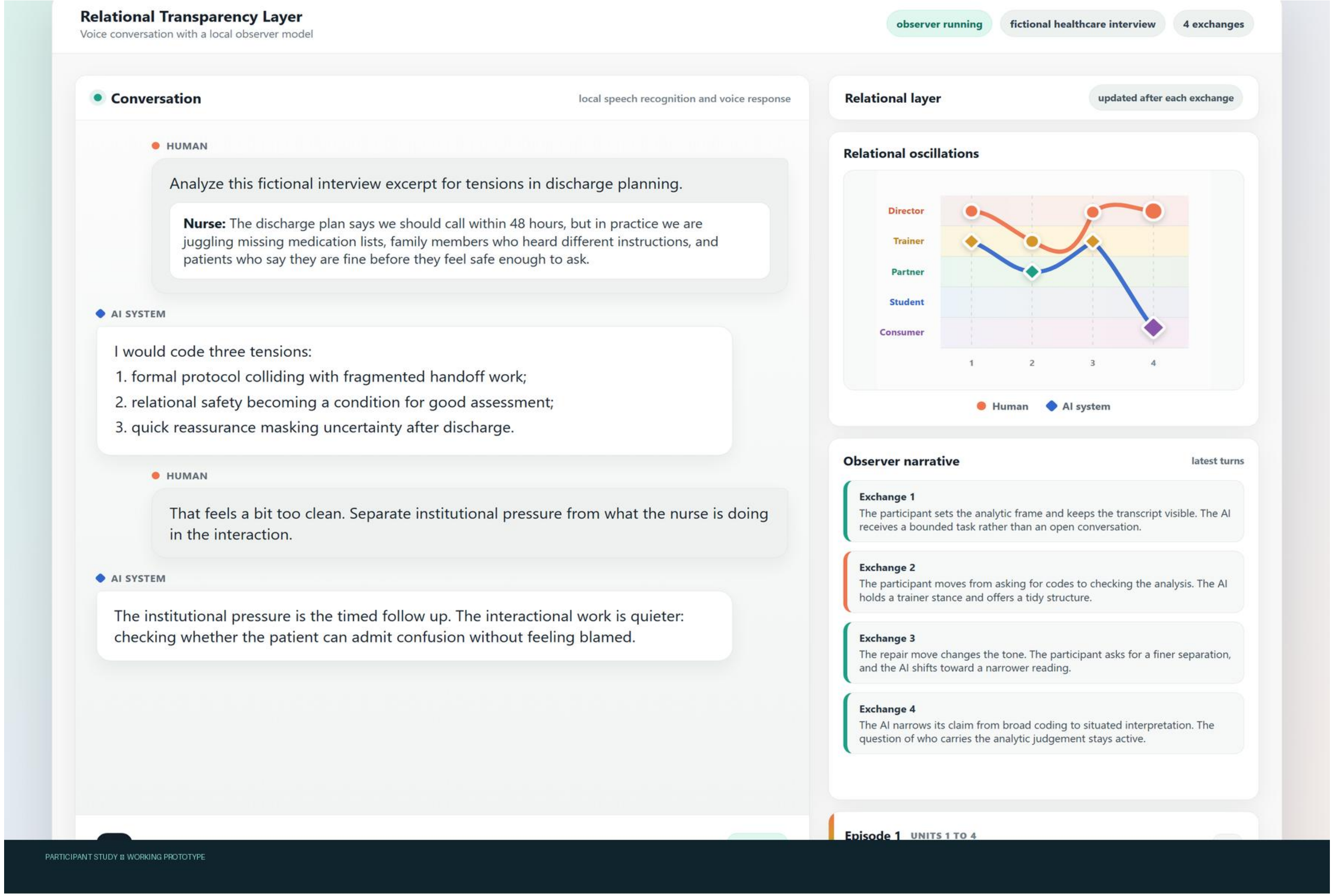
WORKING ARTIFACT

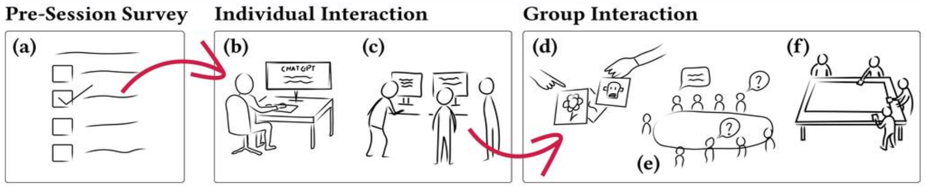
Voice based desktop prototype

STATUS

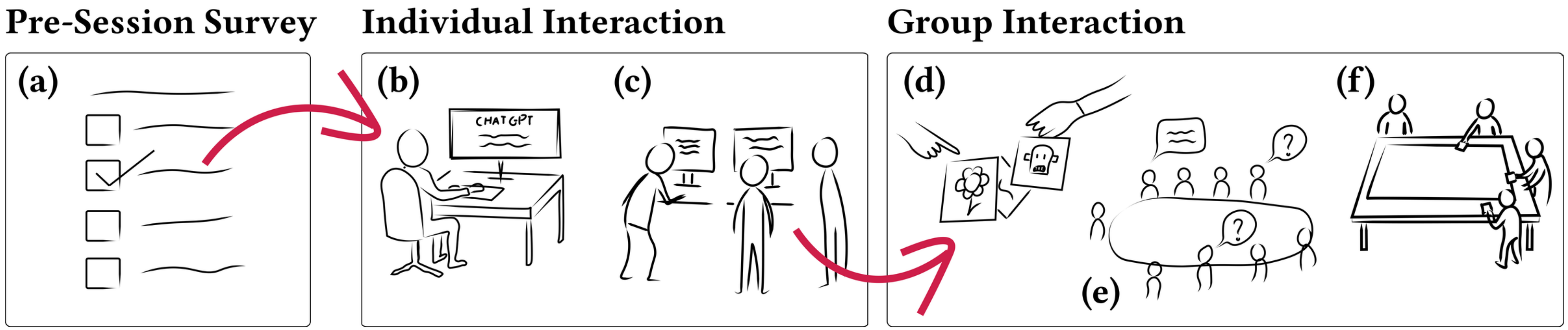
Participant study complete · prototype evaluation pending

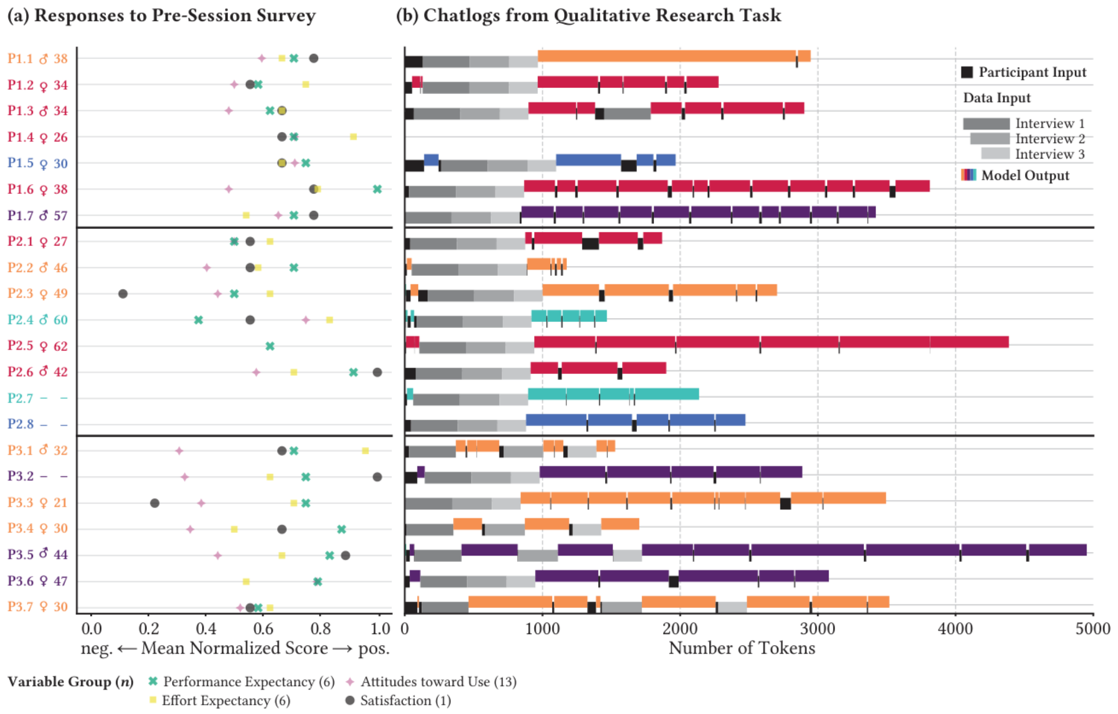
live portfolio | email





How do researchers describe AI, and how does that relation appear in their interaction?





EVIDENCE

DATA

Pre session surveys, chat logs, image choices, roundtable discussion, group worksheets, and observations.

ANALYSIS

One analyst conducted the individual level interpretative phenomenological analysis; another conducted the social level thematic analysis. The team then integrated the findings.

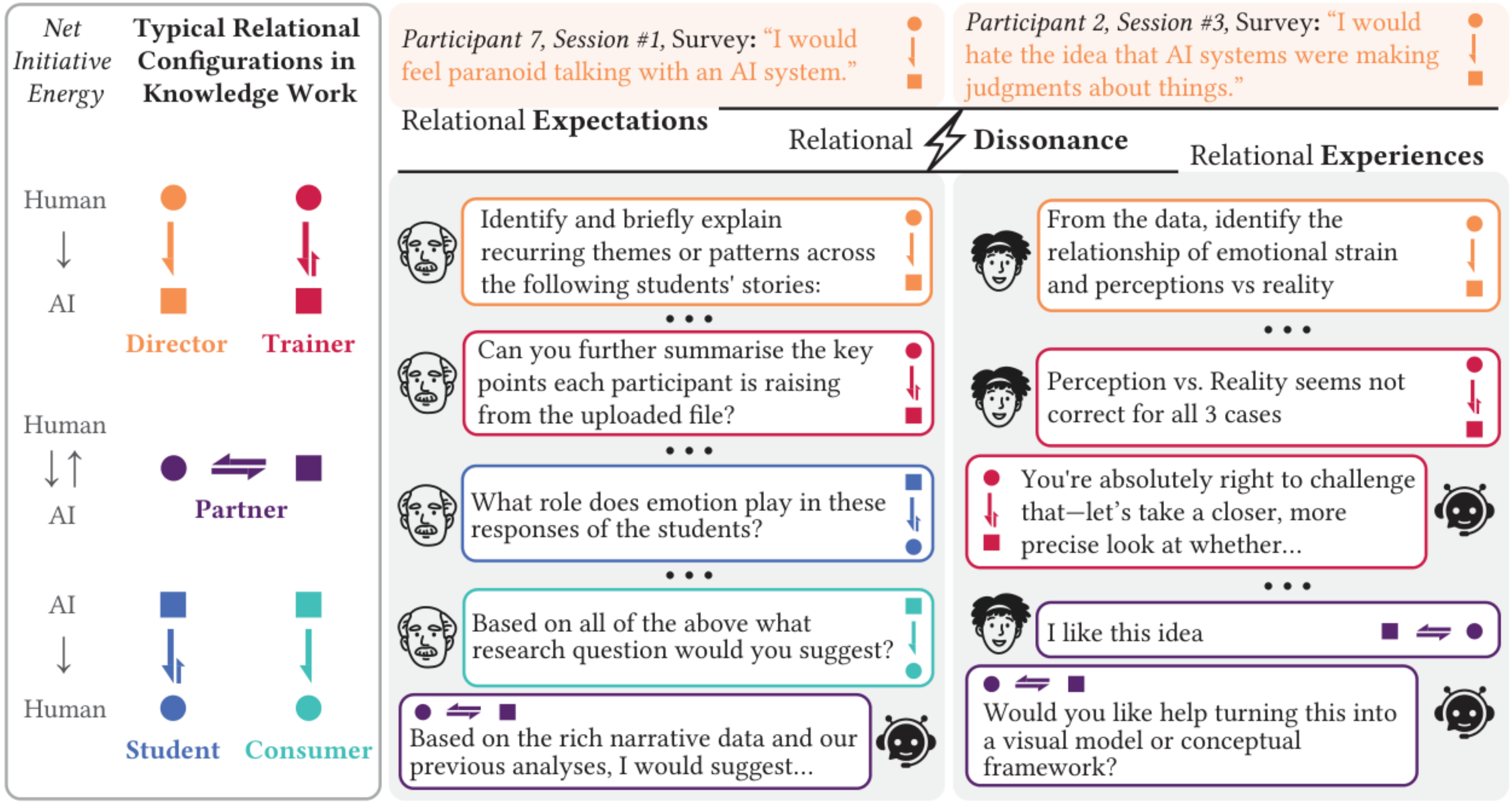
ROLE

First author in a four person interdisciplinary research team.

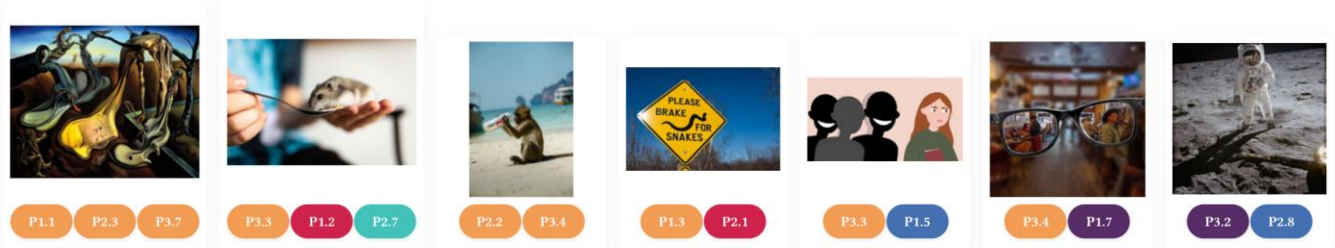
3 workshops 22 participants 4 AI systems

Evidence basis: descriptive survey data, interaction traces, workshop material, and observations.

02 From interaction traces to a design direction Relational Dissonance synthesis



PARTICIPANT ELICITATION ARTIFACTS



SYNTHESIS

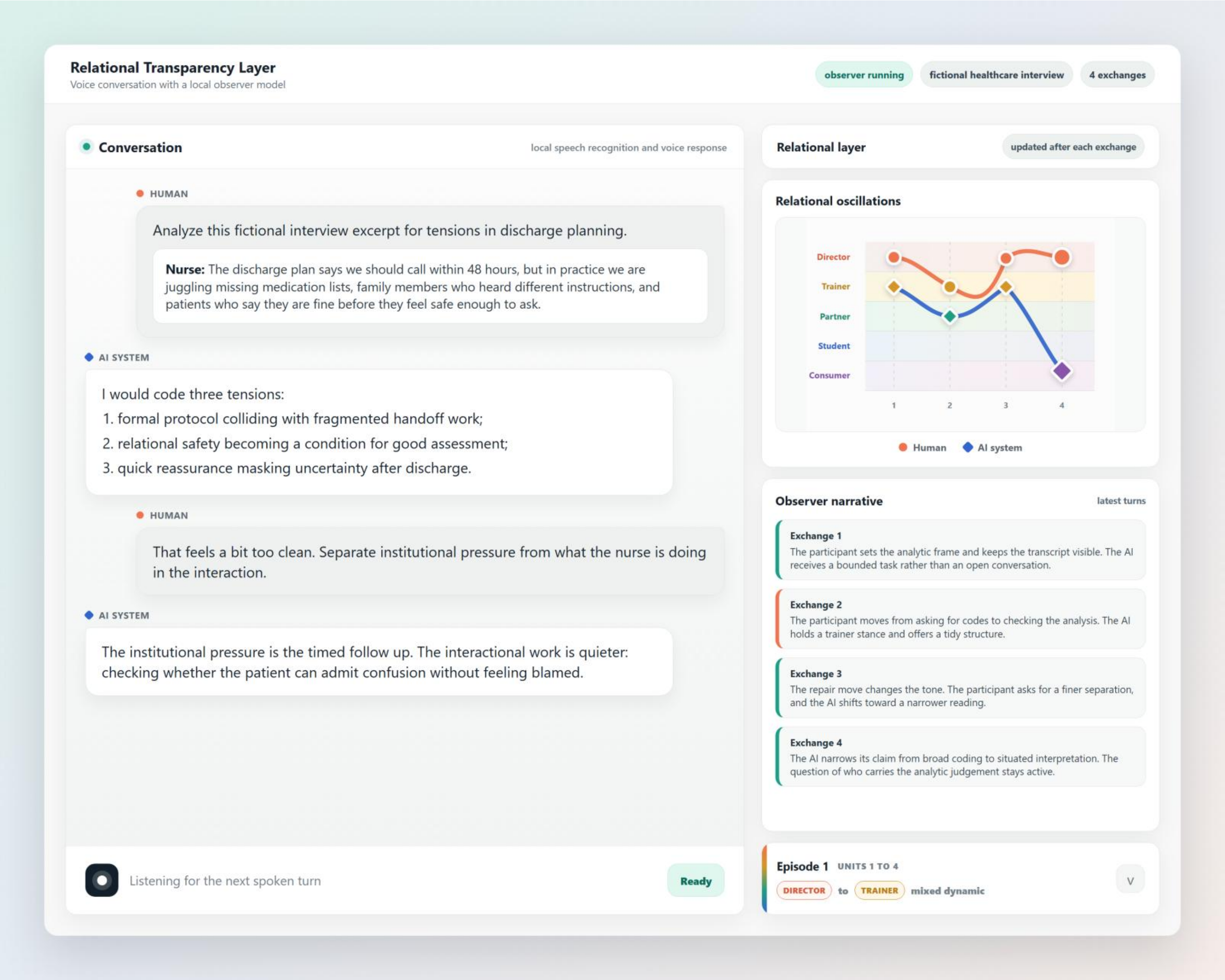
Our analysis found that participants' stated position toward AI often differed from the relation enacted in their chat logs.

Five recurring configurations captured how role, control, and responsibility shifted across the work.

DESIGN RESPONSE

The findings informed Emrecan's later Relational Transparency project and the prototype shown next.

Images in the lower strip were selected and discussed by participants during the workshop.



03 WORKING PROTOTYPE

Relational Transparency Layer

A voice based desktop prototype that places the conversation, a provisional model account, and model generated relational indicators in one inspectable view.

ROLE
Principal investigator and sole demo paper author.

SEPARATE
Conversation and observation run as separate processes.

TRACE
The account is linked to completed turns and visible interaction evidence.

QUESTION
A demo user can compare the observer account with the transcript and their own experience.

BOUNDARY
Working demo. Formal participant evaluation remains future work.

WATCH THE 55 SECOND DEMO

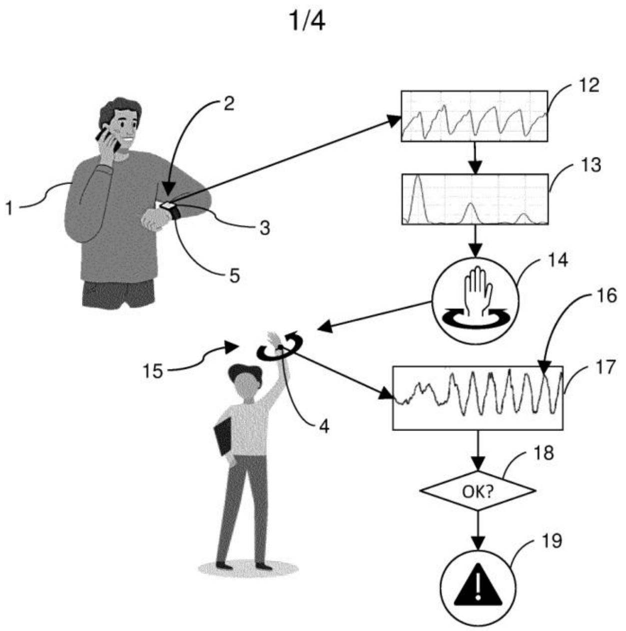


FIG. 1

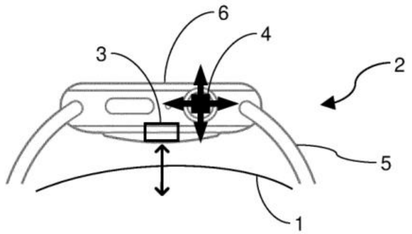


FIG. 2

WO2024235471A1

| guided interaction for wearable sensing accuracy

HUMAN MACHINE INTERACTION

HMI research and technology planning

HMI Technology Planning Specialist and HMI Research Intern at Huawei, 2022 to 2023.

PUBLIC RECORD

Named co inventor on five published international patent applications under the PCT.

- WO2024235471A1 wearable sensing accuracy
- WO2024213244A1 distributed smart devices
- WO2024188442A1 multisensory smart insole
- WO2024179676A1 communication haptics
- WO2024179677A1 activity related haptics

EVIDENCE BOUNDARY

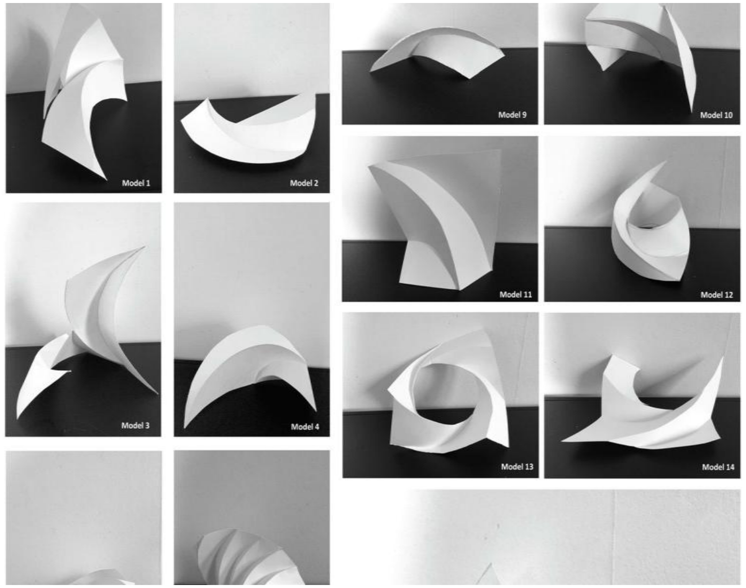
Public records confirm co inventorship and publication of the five PCT applications. Specific research methods, decision authority, implementation, adoption, and product outcomes are unclaimed.

EMPLOYMENT RECORD SCALE (CV)

30+	5	11+	6
technology reports	PCT applications	research institutions	companies

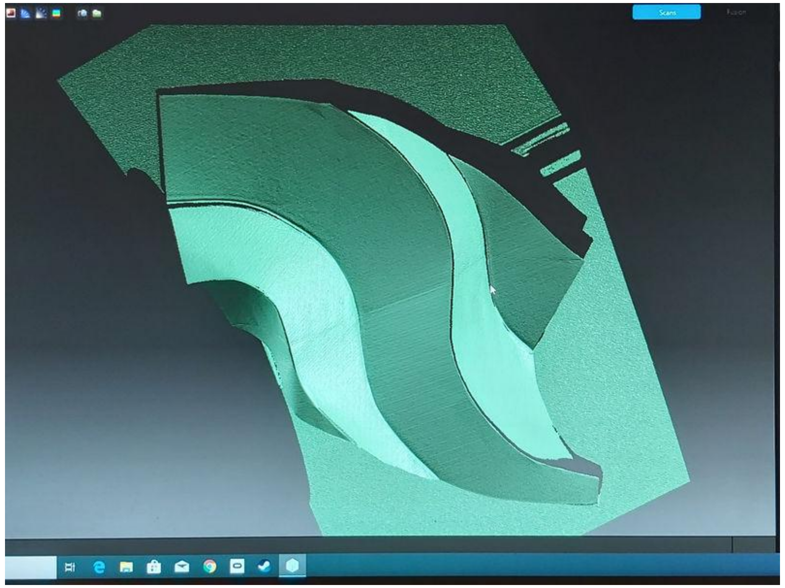
05 Curved Folding A physical to digital research sequence

A design process moved between paper models, scanning, simulation, and plywood fabrication.



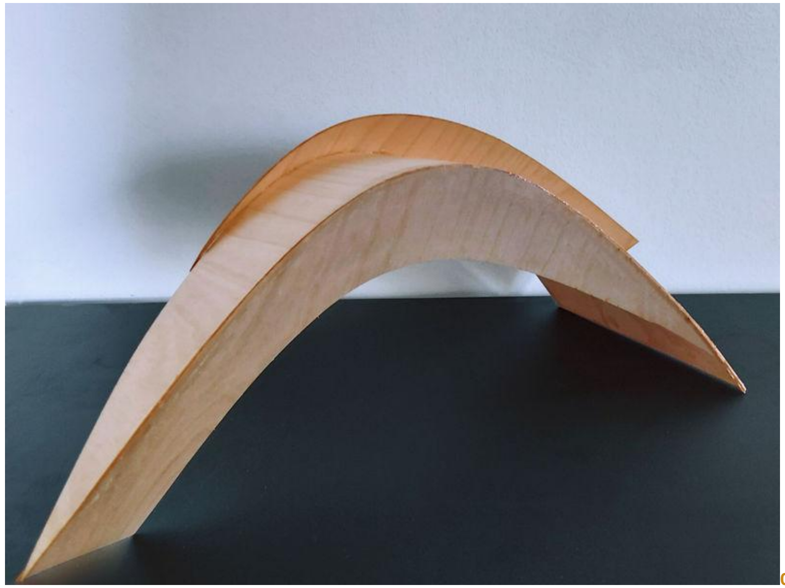
01

Explore and select



02

Translate and test



03

Fabricate and inspect

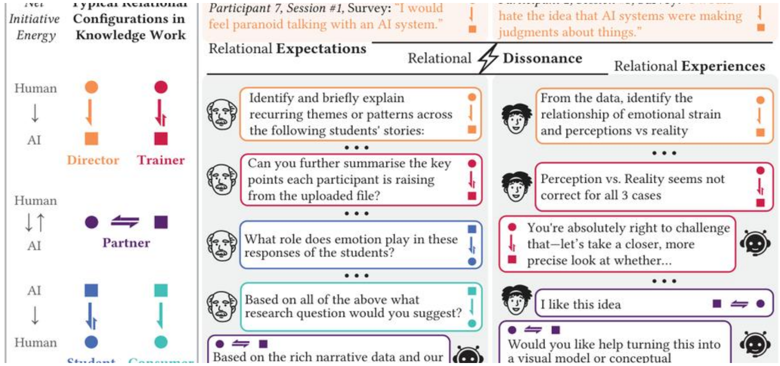
42	5	2	CHI 2021
paper models	shortlisted	plywood prototypes	verified publication

Role: first author; the paper states that the design process was mainly carried out by the first author.

Exploring a Feedback-Oriented Design Process Through Curved Folding | DOI 10.1145/3411764.3445639

What the portfolio demonstrates

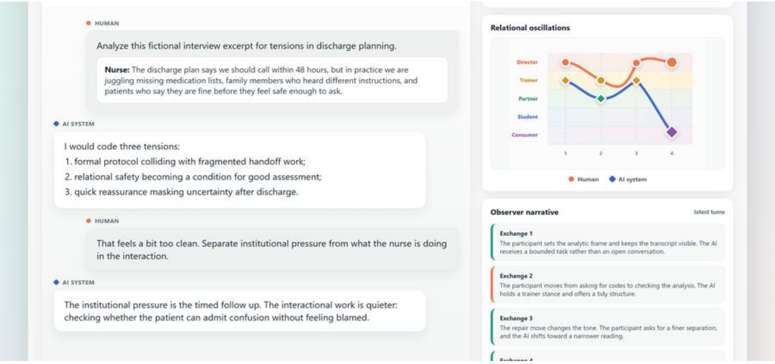
Evidence mapped to Verda



PARTICIPANT RESEARCH

Study and synthesis

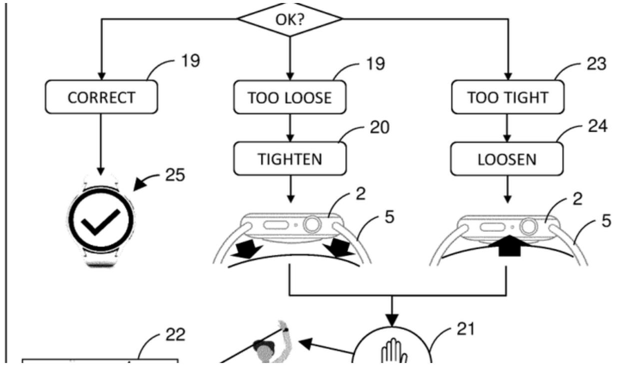
3 workshops · 22 researchers · multiple qualitative evidence sources



PROTOTYPE INQUIRY

System behaviour

Working voice demo · provisional observer account · evaluation pending



INDUSTRY RESEARCH

Public invention record

Huawei · 5 published PCT applications · named co inventor

APPLICATION TO VERDA

At Verda, I would apply this research practice to developer and machine learning infrastructure workflows, studying how users form expectations about agent behaviour, control handoffs, errors, and recovery.

Frame the problem with participants · trace interaction evidence · build artifacts that can be questioned

EMRECAN GULAY

Doctor of Arts · Espoo, Finland

Concurrent Data Agent · Aalto School of Business · research data management and Open Research Network

emrecan.gulay@aalto.fi
Live research portfolio
emrecangulay.com
LinkedIn
Google Scholar

VERIFIED WORK Relational Dissonance · CHI 2026 | Curved Folding · CHI 2021